

Object Detection Metrics: Understanding MAP

In object detection, evaluating a model's performance requires more than simple accuracy. We use a combination of metrics to determine how well the model localizes objects and identifies their classes.

1. The Foundation: IoU (Intersection over Union)

IoU measures the overlap between the **Predicted Bounding Box** and the **Ground Truth Box**. It is calculated as the area of overlap divided by the area of union. An IoU > 0.5 is typically considered a "True Positive".

2. Precision and Recall

- **Precision:** Of all predicted objects, how many were correct? $\text{Precision} = \frac{TP}{TP + FP}$
- **Recall:** Of all actual objects in the image, how many did the model find? $\text{Recall} = \frac{TP}{TP + FN}$

AP (Average Precision): AP is the area under the Precision-Recall curve. It summarizes the model's ability to balance precision and recall across different confidence thresholds for a single class.

3. Mean Average Precision (MAP)

MAP is the average of the AP calculated for all classes in the dataset. It provides a single score to evaluate the entire detection system.

MAP@50 (or mAP@.5)

The MAP calculated at a fixed IoU threshold of 0.50. This is a relatively "lenient" metric that focuses on whether the model found the object, even if the box alignment isn't perfect.

MAP@50-95 (or mAP@.5:.95)

This is the standard COCO metric. It is the average MAP across ten different IoU thresholds: 0.50, 0.55, ..., 0.95. This rewards models that are highly precise in their bounding box localization.

4. Metrics in YOLO (You Only Look Once)

YOLO models optimize for these metrics by predicting bounding boxes and class probabilities simultaneously. During training, the loss function (often a combination of CIoU/DIoU loss and classification loss) directly encourages higher IoU and better class separation, which translates to higher MAP scores during validation.

5. Initial YOLO Results Analysis

Our initial results indicate how the model is performing in its current state. Higher **mAP@50** suggests strong object recognition, while the gap between **mAP@50** and **mAP@50-95** reveals how much room there is to improve the "tightness" of the bounding boxes. If recall is high but precision is low, the model may be overpredicting (too many false positives).

6. REFERENCES :

- **Ultralytics.** *Mean Average Precision (MAP) in Object Detection.*
- **Baeldung.** *Mean Average Precision in Object Detection.*
- **Roboflow.** *Understanding MAP for Object Detection.*

Average Precision (AP) is calculated for each object class by plotting the Precision–Recall (PR) curve at different confidence thresholds. AP is the area under the Precision–Recall curve, which summarizes the model's performance for that single class.

Mean Average Precision (MAP) is then obtained by calculating the average of the AP values across all object classes in the dataset. It provides a single overall score that represents the performance of the object detection model.

Formula:
$$\text{MAP} = \frac{AP_1 + AP_2 + \dots + AP_n}{n}$$

where:

$(AP_1, AP_2, \dots, AP_n)$ are the Average Precision values for each class.

(n) is the total number of object classes. For example, if a dataset has six defect classes, the AP for each class is calculated separately, and their average gives the final MAP score.